

On Erdős Problem 767: Cycles with Chords

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Abstract

For integers $k \geq 1$ and $n \geq k + 2$, let $g_k(n)$ be the maximum number of edges in an n -vertex graph containing no cycle with a vertex incident with at least k chords. Erdős conjectured that $g_k(n) = (k + 1)(n - k - 1)$ for $n \geq 2k + 2$. Lewin found a counterexample. Bollobás later conjectured that there exists a function $n(k)$ such that $g_k(n) = (k + 1)(n - k - 1)$ for all $n \geq n(k)$. Jiang confirmed this by proving the formula for all $n \geq 3k + 3$ when $k \geq 1$.

In this paper, we determine $g_k(n)$ completely. For all $k \geq 1$ and $n \geq k + 2$, we prove

$$g_k(n) = \max \left\{ \left\lfloor \frac{(k+1)n}{2} \right\rfloor, \max_{\substack{a \in \mathbb{Z} \\ \lfloor (k+1)/2 \rfloor + 1 \leq a \leq k+1}} \left\{ a(n-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor \right\} \right\}.$$

For $k \geq 2$, we prove $g_k(n) = (k + 1)(n - k - 1)$ when $n \geq \lceil (5k + 1)/2 \rceil$, and this threshold is sharp. Our proof builds on the method developed by Ma and the second author in [18].

Keywords. chorded cycles; chords; extremal graph theory; stability.

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1 Introduction

All graphs in this paper are finite and simple. A *chord* of a cycle C is an edge joining two nonconsecutive vertices of C . A cycle with a chord is called a *chorded cycle*. We study cycles containing a vertex incident with many chords.

For $k \geq 1$ and $n \geq k + 2$, let $g_k(n)$ denote the maximum number of edges in an n -vertex graph containing no cycle with a vertex incident with at least k chords. The case $k = 1$ follows from a classical theorem of Pósa [19, solution prob, 10.2, p.376]: every graph of order $n \geq 4$ with at least $2n - 3$ edges contains a chorded cycle. The examples K_3 and $K_{2,n-2}$ show that both $n \geq 4$ and $2n - 3$ are sharp. For general k and $n \geq 2k + 2$, Erdős [10] conjectured that $g_k(n) = (k + 1)(n - k - 1)$. This conjecture was listed as Problem #767 on Bloom's Erdős Problems website [2]. Lewin disproved this conjecture; see [4, p. 398, Problem 12]. Bollobás then posed the following problem.

Problem 1.1 (Bollobás [4, p. 398, Problem 13]). *For every integer $k \geq 1$, does there exist an integer $n(k)$ such that $g_k(n) = (k + 1)(n - k - 1)$ for every $n \geq n(k)$?*

Jiang answered Problem 1.1 affirmatively.

Theorem 1.2 (Jiang [17]). *Let $k \geq 1$ and $n \geq 3k + 3$. Then $g_k(n) = (k + 1)(n - k - 1)$.*

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Jiang's theorem leaves the range $k + 2 \leq n \leq 3k + 2$ open. Our main result determines $g_k(n)$ completely.

Theorem 1.3. *Let $k \geq 1$ and $n \geq k + 2$. Then*

$$g_k(n) = \max \left\{ \left\lfloor \frac{(k+1)n}{2} \right\rfloor, \max_{\substack{a \in \mathbb{Z} \\ \lfloor (k+1)/2 \rfloor + 1 \leq a \leq k+1}} \left\{ a(n-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor \right\} \right\}.$$

For $k \geq 2$, Theorem 1.3 implies that $g_k(n) = (k+1)(n-k-1)$ for $n \geq \lceil (5k+1)/2 \rceil$. This threshold is sharp. Indeed, when $n = \lceil (5k+1)/2 \rceil - 1$, we can construct a graph with $(k+1)(n-k-1) + 1$ edges.

The main theorem and its proof are closely related to the work of Ma and Ning on stability on circumference of graphs [18]. Stability results for Kopylov's theorem for graphs with bounded circumference were studied by Füredi, Kostochka, and Verstraëte [14], and were later completed by Füredi, Kostochka, Luo, and Verstraëte [13]. Ma and Ning [18] established a stability version of a conjecture proposed by Woodall in 1976 (see [24]). They also discovered that Woodall's conjecture is related to a theorem on long cycles proved by Bondy in 1971 [3]. As a tool for their proof, they obtained a stability version of Bondy's theorem. Motivated by the idea of Jiang [17] using Bondy's theorem to prove that Problem 1.1 holds for $n \geq 3k + 3$, in this paper, we use Ma and Ning's stability result [18] to handle the remaining range of n .

The study of chords in cycles is also related to several classical extremal and structural problems on cycles. Thomassen [20] conjectured that every longest cycle in a 3-connected graph has a chord. Thomassen [21] first verified the conjecture for several classes of graphs and recently proved that every 3-connected graph contains a longest cycle with a chord [22]. Another related problem concerns nested cycles. Bollobás [5] introduced this problem. Chen, Erdős, and Staton [7] proved that, for every fixed positive integer k , there exists a constant c_k such that every n -vertex graph with at least $c_k n$ edges contains k edge-disjoint nested cycles. Gil Fernández, Kim, Kim, and Liu [15] proved that there exists an absolute constant $C > 0$ such that every n -vertex graph with at least Cn edges contains two edge-disjoint nested cycles without crossings. Chakraborti, Janzer, Methuku, and Montgomery [6] proved that, for some absolute constant t , $O_k(n(\log n)^t)$ edges are sufficient to guarantee k pairwise edge-disjoint cycles with the same vertex set. For further results on chorded and nested cycles, see [1, 16, 23].

The paper is organized as follows. We begin with a brief proof sketch in Section 2. In Section 3, we present the extremal constructions and establish several estimates for the upper bound. In Section 4, we collect the closure, switching, and stability tools. Finally, we prove Theorem 1.3 in Section 5.

2 Notation and proof sketch

For $X \subseteq V(G)$, let $G[X]$ denote the subgraph induced by X and let $G - X := G[V(G) \setminus X]$. For $v \in V(G)$, let $G - v := G - \{v\}$. If H is a subgraph of G , let $G - H := G - V(H)$. For disjoint sets $X, Y \subseteq V(G)$, let $E_G(X, Y)$ be the set of edges with one end in X and the other in Y , and set $e_G(X, Y) := |E_G(X, Y)|$. For two disjoint subgraphs H_1 and H_2 of G , let $e_G(H_1, H_2) := e_G(V(H_1), V(H_2))$. Let $e(G) := |E(G)|$.

For each $v \in V(G)$, let $N_G(v)$ denote the neighborhood of v , set $d_G(v) := |N_G(v)|$, and let $\delta(G) := \min_{v \in V(G)} d_G(v)$. For $X \subseteq V(G)$, let $N_G(X) := \{v \in V(G) \setminus X : N_G(v) \cap X \neq \emptyset\}$. For a subgraph H of G and a cycle C in G , set $N_C(H) := N_G(V(H)) \cap V(C)$. For $x \in V(G)$, set $N_H(x) := N_G(x) \cap V(H)$.

The length of a path or cycle is its number of edges. For a path P and vertices $u, v \in V(P)$, let uPv be the subpath of P from u to v , including both endpoints. For a subgraph H of G , an (x, H, y) -path is an x - y path whose internal vertices lie in $V(H)$. An (x, Z, y) -path

is defined similarly for a vertex set Z . Let $\omega(G)$ be the clique number and $c(G)$ for the circumference of G . When we say that C is oriented, we mean that it is oriented clockwise or counterclockwise. If C is oriented, then v^+ and v^- denote the successor and predecessor of v on C . If C is oriented and $u, v \in V(C)$, then uCv denotes the directed u - v segment of C . A cycle C in G is *locally maximal* if there is no cycle C' with $|E(C')| > |E(C)|$ that uses at most two edges of $E_G(V(C), V(G) \setminus V(C))$. Every longest cycle is locally maximal. A graph is Hamiltonian-connected if every two distinct vertices are joined by a Hamilton path.

For an integer r , the r -closure of G is obtained by repeatedly joining two nonadjacent vertices whose degree sum is at least r . The resulting graph is unique [3, 8]. Thus G is r -closed if $d_G(x) + d_G(y) < r$ for every pair of nonadjacent vertices x, y . If C is a cycle of length c , the C -closure of G , denoted by $\text{cl}_C(G)$, is obtained by replacing $G[V(C)]$ with its $(c+1)$ -closure and leaving all other adjacencies unchanged.

Proof sketch. The case $k = 1$ follows from Pósa's theorem, so we may assume that $k \geq 2$. The lower bound follows from the two constructions in Section 3: a nearly $(k+1)$ -regular construction and a join-type construction. For the upper bound, we first establish it for $k+2 \leq n \leq \lceil (5k+1)/2 \rceil$ and then extend the bound to all n by induction.

First, suppose that $k+2 \leq n \leq \lceil (5k+1)/2 \rceil$. We argue by contradiction and choose a minimal counterexample G . Then G is 2-connected and non-Hamiltonian. Let C be a longest cycle of G , and set $c = |E(C)|$. Applying Lemma 4.4 to G and C , we obtain two cases for the subgraph induced by $V(C)$ in the C -closure of G : either deleting $s-1$ vertices of degree at most s leaves a clique, or there are $\lfloor c/2 \rfloor - 1$ vertices of degree at most $\lfloor c/2 \rfloor$. These two cases give bounds on $e(G[V(C)])$ and on the number of edges with at least one endpoint outside C .

Next, we first treat the case $c \leq 9$ separately, and then suppose that $c \geq 10$. If $e(G-C) + e_G(G-C, C) \leq (\lfloor c/2 \rfloor - 1)(n-c)$, then combining this inequality with the degree bounds on C gives the bound in Theorem 1.3. Otherwise, the Ma-Ning stability theorem [18] (see Theorem 4.2) implies that G belongs to one of the exceptional families or $G \subseteq W_{n, \lfloor c/2 \rfloor, c}$, where the definition of $W_{n, \lfloor c/2 \rfloor, c}$ is postponed to Section 4. We also show that if G belongs to one of the exceptional families, then $G \subseteq W_{n, \lfloor c/2 \rfloor, c}$. Finally, the structure of $W_{n, \lfloor c/2 \rfloor, c}$ yields the bound in Theorem 1.3.

For $n \geq \lceil (5k+1)/2 \rceil$, the upper bound follows by induction on n , using an endpoint v of a longest path with $d_G(v) \leq k+1$.

3 Extremal constructions and arithmetic

For $\ell \geq 2$, an ℓ -path-fan in G consists of a vertex x and a path P in $G-x$ such that $|N_G(x) \cap V(P)| \geq \ell$. We call x the center of the path-fan.

Lemma 3.1. *For $k \geq 1$, a graph contains a $(k+2)$ -path-fan if and only if it contains a cycle with a vertex incident with at least k chords.*

Proof. First, suppose that a vertex x is incident with at least k chords of a cycle C . Then x has at least $k+2$ neighbors on the path $C-x$. Hence x and $C-x$ form a $(k+2)$ -path-fan.

Conversely, suppose that x and a path P in $G-x$ form a $(k+2)$ -path-fan. Let uPw be the minimal subpath of P containing $N_G(x) \cap V(P)$. In particular, both u and w are adjacent to x . The cycle $xuPwx$ contains all neighbors of x on uPw . Thus, x is incident with at least k chords of this cycle. $\square$

For every integer $k \geq 2$, set

$$I_k := \left\{ a \in \mathbb{Z} : \left\lfloor \frac{k+1}{2} \right\rfloor + 1 \leq a \leq k+1 \right\}.$$

For an integer $n \geq 1$, define

$$R_k(n) := \left\lfloor \frac{(k+1)n}{2} \right\rfloor, \quad J_k(n) := \max_{a \in I_k} a(n-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor.$$

Finally, set

$$F_k(n) := \begin{cases} \binom{n}{2}, & 1 \leq n \leq k+1, \\ \max\{R_k(n), J_k(n)\}, & n \geq k+2. \end{cases}$$

3.1 The lower-bound constructions

We give two families of graphs containing no $(k+2)$ -path-fan: a nearly regular construction with $R_k(n)$ edges and a join-type construction with $a(n-a) + \lfloor a(k+1-a)/2 \rfloor$ edges for each $a \in I_k$.

Construction 3.2. Take the cyclic group $\mathbb{Z}_n$ where $n \geq k+2$. Define edge set $M_1 = \{\{i, i+j\} : i \in \mathbb{Z}_n, 1 \leq j \leq \lfloor (k+1)/2 \rfloor\}$. Moreover, if $k+1$ is odd, $M_2 = \{\{i, i + \lfloor n/2 \rfloor\} : 0 \leq i \leq \lfloor n/2 \rfloor - 1\}$, where all additions are taken in $\mathbb{Z}_n$. Then define the graph G as follows:

- if $k+1$ is even, let $V(G) = \mathbb{Z}_n$, $E(G) = M_1$;
- if $k+1$ is odd, let $V(G) = \mathbb{Z}_n$, $E(G) = M_1 \cup M_2$.

If $k+1$ is even, then G is $(k+1)$ -regular, and $e(G) = (k+1)n/2 = R_k(n)$. If $k+1$ is odd, then $(\mathbb{Z}_n, M_1)$ is k -regular. Since $n \geq k+2$, we have $\lfloor n/2 \rfloor > \lfloor (k+1)/2 \rfloor$. Then M_2 is disjoint from M_1 and forms a matching of size $\lfloor n/2 \rfloor$. Thus the maximum degree of G is at most $k+1$ and $e(G) = kn/2 + \lfloor n/2 \rfloor = \lfloor (k+1)n/2 \rfloor = R_k(n)$. Since the center of a $(k+2)$ -path-fan has at least $k+2$ distinct neighbors, every graph with maximum degree at most $k+1$ contains no $(k+2)$ -path-fan; see Figure 1.

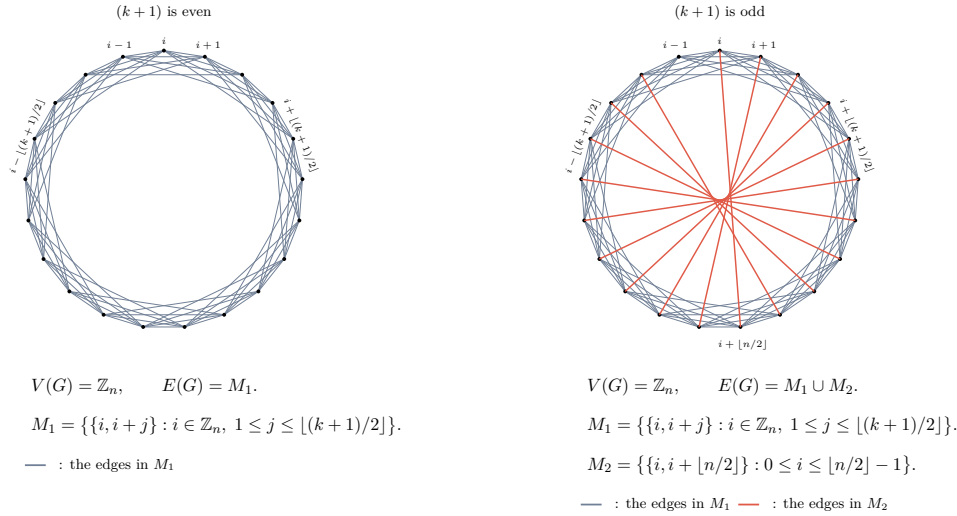


Figure 1: The nearly regular construction.

Construction 3.3. Let $a \in I_k$. Partition the vertex set into X and Y with $|X| = a$ and $|Y| = n-a$, where Y is an independent set. Identify X with the cyclic group $\mathbb{Z}_a$. Define two edge sets $M_0 = \{\{x, y\} : x \in X, y \in Y\}$, and $M_1 = \{\{i, i+j\} : i \in \mathbb{Z}_a, 1 \leq j \leq \lfloor (k+1-a)/2 \rfloor\}$. Moreover, if $k+1-a$ is odd, define $M_2 = \{\{i, i + \lfloor a/2 \rfloor\} : 0 \leq i \leq \lfloor a/2 \rfloor - 1\}$, where all additions are taken in $\mathbb{Z}_a$. Then we define the graph G as follows:

- if $k+1-a$ is even, let $V(G) = X \cup Y$, $E(G) = M_0 \cup M_1$;
- if $k+1-a$ is odd, let $V(G) = X \cup Y$, $E(G) = M_0 \cup M_1 \cup M_2$.

The resulting graph G has $e(G) = a(n - a) + \lfloor a(k + 1 - a)/2 \rfloor$ edges and contains no $(k + 2)$ -path-fan. Indeed, if the center of a $(k + 2)$ -path-fan is a vertex $y \in Y$, then $N_G(y) = X$. Hence $d_G(y) = a \leq k + 1$. If the center is a vertex $x \in X$, let P be any path in $G - x$. Since Y is independent, P contains at most a vertices of Y . Moreover, x has at most $k + 1 - a$ neighbors in X . Therefore, $|N_G(x) \cap V(P)| \leq a + (k + 1 - a) = k + 1$. Thus G contains no $(k + 2)$ -path-fan; see Figure 2.

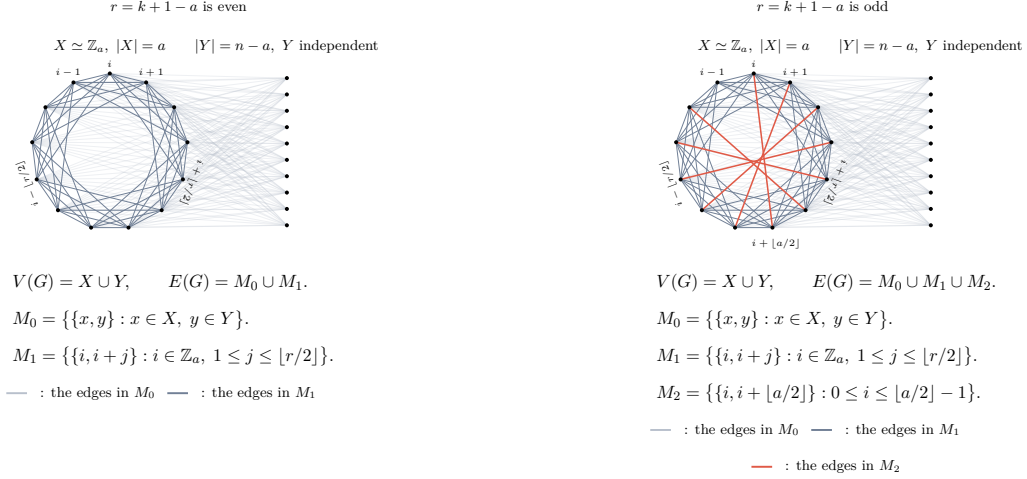


Figure 2: The join-type construction.

The two constructions show $g_k(n) \geq \max\{R_k(n), J_k(n)\} = F_k(n)$ for all $n \geq k + 2$. Moreover, Construction 3.3 shows that the threshold $n = \lceil (5k + 1)/2 \rceil$ is sharp.

Remark 3.4. Let $k \geq 2$ and $n = \lceil (5k + 1)/2 \rceil - 1$. Construction 3.3 with $a = k$ gives a graph containing no $(k + 2)$ -path-fan and having $k(n - k) + \lfloor k/2 \rfloor$ edges, where $k(n - k) + \lfloor k/2 \rfloor - (k + 1)(n - k - 1) = 1$. Thus the threshold $n = \lceil (5k + 1)/2 \rceil$ is best possible.

3.2 Estimates for the upper bound

We next prove several lemmas. These lemmas will be used in the calculations in Section 5 and in the induction. Readers who are mainly interested in the proof of Theorem 1.3 can skip these lemmas and go directly to the main proof in Section 5 to understand the overall structure. They can also use a computer program to check the calculations independently.

Recall the functions R_k, J_k, F_k defined in the beginning of Section 3. The first lemma determines when $J_k(n)$ exceeds $R_k(n)$ and will be used to derive the piecewise formula for $F_k(n)$.

Lemma 3.5. *Let $k \geq 2$ and let $N_k \geq k + 2$ be the smallest integer such that $J_k(N_k) > R_k(N_k)$. Then $J_k(m) > R_k(m)$ for all $m \geq N_k$. Moreover,*

- (i) $\left\lfloor \left(1 + \frac{\sqrt{3}}{2}\right)(k + 1) \right\rfloor + 1 \leq N_k \leq \left\lceil \left(1 + \frac{\sqrt{3}}{2}\right)(k + 1) \right\rceil + 1.$
- (ii) *if $J_k(n) \leq R_k(n)$ for some $n \geq k + 2$, then $n \leq \left\lceil \left(1 + \frac{\sqrt{3}}{2}\right)(k + 1) \right\rceil \leq 2k + 2.$*

Proof. For $k = 2$, we have $R_2(n) = \lfloor 3n/2 \rfloor$ and $J_2(n) = \max\{2n - 3, 3n - 9\}$. Thus, $N_2 = 7$, and both assertions follow. We may therefore assume that $k \geq 3$. The integer N_k exists because $(k + 1)(n - k - 1) > R_k(n)$ for all sufficiently large n .

Let $a \in I_k$ be such that $J_k(N_k) = a(N_k - a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor$. For any $m > N_k$, we have

$$\begin{aligned} J_k(m) &\geq a(m-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor = a(N_k-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor + a(m-N_k) \\ &= J_k(N_k) + a(m-N_k), \\ R_k(m) &= \left\lfloor \frac{(k+1)m}{2} \right\rfloor \leq \left\lfloor \frac{(k+1)N_k}{2} \right\rfloor + a(m-N_k) = R_k(N_k) + a(m-N_k). \end{aligned}$$

Since $J_k(N_k) > R_k(N_k)$, we obtain $J_k(m) > R_k(m)$ for all $m \geq N_k$. We prove the remaining assertion by constructing a function $\Phi(x)$ and dividing n into two ranges according to the minimum value of $\Phi(x)$.

(i) Fix $a \in I_k$. We first compare $a(n-a) + \frac{a(k+1-a)}{2}$ and $\frac{(k+1)n}{2}$. Let $x := 2a - k - 1$ and define $\Phi(x) := \frac{(k+1)^2}{4x} + (k+1) + \frac{3x}{4}$. Then

$$a(n-a) + \frac{a(k+1-a)}{2} - \frac{(k+1)n}{2} = \frac{x}{2}(n - \Phi(x)). \quad (3.1)$$

The function Φ attains its minimum at $x_0 = \frac{k+1}{\sqrt{3}}$, where $\Phi(x_0) = \left(1 + \frac{\sqrt{3}}{2}\right)(k+1)$.

Case 1: $n \leq \lfloor \Phi(x_0) \rfloor$.

For every $a \in I_k$, we have $n \leq \Phi(x_0) \leq \Phi(x)$. From (3.1), $a(n-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor \leq \left\lfloor \frac{(k+1)n}{2} \right\rfloor$ for all $a \in I_k$. Hence $J_k(n) \leq R_k(n)$ when $k+2 \leq n \leq \lfloor \Phi(x_0) \rfloor$.

Case 2: $n \geq \lceil \Phi(x_0) \rceil + 1$.

Set $N = \lceil \Phi(x_0) \rceil + 1$. Choose an integer x such that $x \equiv k+1 \pmod{2}$ and $|x - x_0| \leq 1$. Such a choice is possible because $k+1 \geq 4$. Define $a = \frac{k+1+x}{2}$. Then a is an integer and $\left\lfloor \frac{k+1}{2} \right\rfloor + 1 \leq a \leq k+1$. Hence $a \in I_k$. Since $|x - x_0| < 1$ and $x \geq 2$, we have $0 \leq \Phi(x) - \Phi(x_0) = \frac{3}{4} \frac{(x-x_0)^2}{x} < \frac{3}{4x} \leq \frac{3}{8}$. Consequently, $N - \Phi(x) = (N - \Phi(x_0)) - (\Phi(x) - \Phi(x_0)) > 1 - \frac{3}{8} = \frac{5}{8}$. From (3.1), we have $a(N-a) + \frac{a(k+1-a)}{2} - \frac{(k+1)N}{2} = \frac{x}{2}(N - \Phi(x)) > \frac{1}{2}$. Then,

$$J_k(N) \geq a(N-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor \geq a(N-a) + \frac{a(k+1-a)}{2} - \frac{1}{2} > \frac{(k+1)N}{2} \geq R_k(N).$$

Therefore $J_k(N) > R_k(N)$.

Since $J_k(n) \leq R_k(n)$ for all $n \leq \lfloor \Phi(x_0) \rfloor$ by Case 1, we have $N_k \geq \lfloor \Phi(x_0) \rfloor + 1$. Since $J_k(N) > R_k(N)$ with $N = \lceil \Phi(x_0) \rceil + 1$, we have $J_k(m) > R_k(m)$ for every $m \geq N$. Therefore, $N_k \leq N = \lceil \Phi(x_0) \rceil + 1$. Combining the two bounds, we obtain

$$\left\lfloor \left(1 + \frac{\sqrt{3}}{2}\right)(k+1) \right\rfloor + 1 \leq N_k \leq \left\lceil \left(1 + \frac{\sqrt{3}}{2}\right)(k+1) \right\rceil + 1.$$

(ii) If $n \geq k+2$ and $J_k(n) \leq R_k(n)$, then $n < N_k$. Consequently, we have $n \leq N_k - 1 \leq \lceil \Phi(x_0) \rceil$. Since $\Phi(x_0) < 2(k+1)$, $\lceil \Phi(x_0) \rceil \leq 2k+2$. This completes the proof. $\blacksquare$

The following lemma shows that $J_k(n)$ is a linear function of n when $n \geq \lceil (5k+1)/2 \rceil$.

Lemma 3.6. *Let $k \geq 2$. If $n \geq \lceil (5k+1)/2 \rceil$, then $J_k(n) = (k+1)(n-k-1)$.*

Proof. It suffices to show that $a(n-a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor \leq (k+1)(n-k-1)$ for every $a \in I_k$. For $a = k+1$, equality holds. Let $a = k+1-d$, where $d \in \mathbb{Z}$ and $1 \leq d < \frac{k+1}{2}$. Then we have

$$\begin{aligned} &(k+1)(n-k-1) - \left((k+1-d)(n-k-1+d) + \left\lfloor \frac{d(k+1-d)}{2} \right\rfloor \right) \\ &= d(n-2k-2+d) - \left\lfloor \frac{d(k+1-d)}{2} \right\rfloor. \end{aligned} \quad (3.2)$$

For $d = 1$, $d(n - 2k - 2 + d) - \left\lfloor \frac{d(k+1-d)}{2} \right\rfloor = n - 2k - 1 - \lfloor k/2 \rfloor \geq 0$. For $d \geq 2$, $d(n - 2k - 2 + d) - \left\lfloor \frac{d(k+1-d)}{2} \right\rfloor \geq \frac{d}{2}(2n - 5k + 1) \geq 0$. Thus, $J_k(n) = (k+1)(n - k - 1)$ for $n \geq \lceil (5k+1)/2 \rceil$. ■

Next, we give a piecewise formula for $F_k(n)$.

Lemma 3.7. *Let $k \geq 2$ and $n \geq k + 2$. Then*

$$F_k(n) = \begin{cases} \left\lfloor \frac{(k+1)n}{2} \right\rfloor, & k+2 \leq n \leq \left\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \right\rfloor, \\ \left\lfloor \frac{(2n+k+1)^2}{24} \right\rfloor, & \left\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \right\rfloor < n < \left\lceil \frac{5k+1}{2} \right\rceil, \\ (k+1)(n - k - 1), & n \geq \left\lceil \frac{5k+1}{2} \right\rceil. \end{cases}$$

Proof. First suppose that $k+2 \leq n \leq \left\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \right\rfloor$. By Lemma 3.5, the smallest integer $N_k \geq k+2$ for which $J_k(N_k) > R_k(N_k)$ satisfies $N_k \geq \lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \rfloor + 1$. Hence $J_k(n) \leq R_k(n)$. Therefore $F_k(n) = R_k(n)$.

Next suppose that $\left\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \right\rfloor < n < \lceil (5k+1)/2 \rceil$. For $k \in \{2, 3\}$, this interval contains no integer. Then we may assume that $k \geq 4$. Let $a_0 \in I_k$ be such that $J_k(n) = a_0(n - a_0) + \left\lfloor \frac{a_0(k+1-a_0)}{2} \right\rfloor$. Then

$$J_k(n) \leq a_0(n - a_0) + \frac{a_0(k+1-a_0)}{2} = \frac{(2n+k+1)^2}{24} - \frac{3}{2} \left(a_0 - \frac{2n+k+1}{6} \right)^2 \leq \frac{(2n+k+1)^2}{24}.$$

Consequently $J_k(n) \leq \left\lfloor \frac{(2n+k+1)^2}{24} \right\rfloor$.

Since $\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \rfloor < n < \lceil \frac{5k+1}{2} \rceil$, every integer closest to $\frac{(2n+k+1)}{6}$ is in I_k . Choose an integer $a \in I_k$ nearest to $\frac{2n+k+1}{6}$, and set $\epsilon := |a - \frac{2n+k+1}{6}|$. Then $\epsilon \in \left\{0, \frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right\}$ and

$$J_k(n) \geq a(n - a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor = \left\lfloor \frac{(2n+k+1)^2}{24} - \frac{3}{2}\epsilon^2 \right\rfloor.$$

In the following, we divide the proof into four cases according to the value of ϵ . If $\epsilon = 0$, then $\frac{3}{2}\epsilon^2 = 0$. If $\epsilon = \frac{1}{6}$, then $\frac{3}{2}\epsilon^2 = \frac{1}{24}$ and the fractional part of $\frac{(2n+k+1)^2}{24}$ is $\frac{1}{24}$. If $\epsilon = \frac{1}{3}$, then $\frac{3}{2}\epsilon^2 = \frac{1}{6}$ and the fractional part of $\frac{(2n+k+1)^2}{24}$ is $\frac{1}{6}$ or $\frac{2}{3}$. If $\epsilon = \frac{1}{2}$, then $\frac{3}{2}\epsilon^2 = \frac{3}{8}$ and the fractional part of $\frac{(2n+k+1)^2}{24}$ is $\frac{3}{8}$.

Therefore, $\left\lfloor \frac{(2n+k+1)^2}{24} - \frac{3}{2}\epsilon^2 \right\rfloor = \left\lfloor \frac{(2n+k+1)^2}{24} \right\rfloor$. Thus $J_k(n) = \left\lfloor \frac{(2n+k+1)^2}{24} \right\rfloor$. Moreover, $n > (1 + \frac{\sqrt{3}}{2})(k+1)$, and hence $\frac{(2n+k+1)^2}{24} > \frac{(k+1)n}{2}$. It follows that $J_k(n) \geq R_k(n)$. Hence $F_k(n) = J_k(n)$.

Finally, suppose that $n \geq \lceil (5k+1)/2 \rceil$. By Lemma 3.6, we have $J_k(n) = (k+1)(n - k - 1)$. Since $n \geq \lceil (5k+1)/2 \rceil$, we have $n \geq 2k+2$ for $k \geq 2$. Consequently $J_k(n) \geq R_k(n)$, and hence $F_k(n) = J_k(n) = (k+1)(n - k - 1)$. ■

To bound the number of edges of G , we decompose G into edge-disjoint subgraphs. The following inequalities combine the bounds for edge-disjoint subgraphs of G .

Lemma 3.8. *Let $k \geq 2$. If $k+2 \leq n \leq \lceil (5k+1)/2 \rceil$ and $J_k(n) > R_k(n)$, then the following inequalities hold.*

- (i) *For all positive integers x and y satisfying $x + y - 1 \leq n$, $F_k(x) + F_k(y) \leq F_k(x + y - 1)$;*
- (ii) *For all positive integers x and y satisfying $x + y \leq n$, $F_k(x) + F_k(y) \leq F_k(x + y)$.*

Proof. When $k = 2$, the hypothesis $J_k(n) > R_k(n)$ does not hold for $4 \leq n \leq 6$. Thus we may assume that $k \geq 3$.

(i) For $m \geq k + 2$, by definition of F_k , $F_k(m) = \max\{R_k(m), J_k(m)\} \geq J_k(m)$. Hence for $k + 2 \leq m \leq n$, we have either $F_k(m) = J_k(m)$ or $F_k(m) > J_k(m)$.

Claim 3.1. *For every integer m with $k + 2 \leq m \leq n$ and every $a \in I_k$, if $F_k(m) > J_k(m)$, then $F_k(m) \leq a(m - 1)$.*

Proof. Since $F_k(m) > J_k(m)$, we have $F_k(m) = R_k(m)$. Since $m \geq k + 2$, we have $\frac{(k+1)m}{2} \leq \frac{k+2}{2}(m - 1) \leq \left(\left\lfloor \frac{k+1}{2} \right\rfloor + 1\right)(m - 1) \leq a(m - 1)$. Hence $F_k(m) = R_k(m) = \left\lfloor \frac{(k+1)m}{2} \right\rfloor \leq a(m - 1)$. $\square$

Case 1. $1 \leq x, y \leq k + 1$.

If $x + y - 1 \leq k + 1$, then $F_k(x) + F_k(y) = \binom{x}{2} + \binom{y}{2} \leq \binom{x+y-1}{2} = F_k(x + y - 1)$. If $x + y - 1 \geq k + 2$, by symmetry we may assume $x \geq y$. If $\min\{x, y\} = 1$, then the assertion is immediate. We may therefore assume that $2 \leq x, y \leq k + 1$. Since $x + y$ is fixed, $\binom{x}{2} + \binom{y}{2}$ is maximized at $x = k + 1$ and $y = x + y - k - 1$. Hence $\binom{x}{2} + \binom{y}{2} \leq \binom{k+1}{2} + \binom{x+y-k-1}{2}$. Moreover, $\frac{(k+1)(x+y-1)}{2} - \binom{k+1}{2} - \binom{x+y-k-1}{2} = \frac{(x+y-k-1)(2k+3-x-y)}{2} \geq 0$. Therefore $\binom{k+1}{2} + \binom{x+y-k-1}{2} \leq R_k(x + y - 1)$. Consequently, $F_k(x) + F_k(y) = \binom{x}{2} + \binom{y}{2} \leq R_k(x + y - 1) \leq F_k(x + y - 1)$.

Case 2. $1 \leq x \leq k + 1$ and $y \geq k + 2$.

Then $x + y - 1 \geq k + 2$. If $F_k(y) = J_k(y)$, let $b \in I_k$ be the integer such that $J_k(y) = b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor$. Thus,

$$\begin{aligned} F_k(x) + F_k(y) &= \binom{x}{2} + J_k(y) = \binom{x}{2} + b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor \\ &\leq b(x - 1) + b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor = b(x + y - 1 - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor \\ &\leq J_k(x + y - 1) \leq F_k(x + y - 1). \end{aligned}$$

If $F_k(y) = R_k(y)$, then

$$\begin{aligned} F_k(x) + F_k(y) &\leq \binom{x}{2} + \left\lfloor \frac{(k+1)y}{2} \right\rfloor \leq \left\lfloor \frac{(k+1)(x+y-1)}{2} \right\rfloor \\ &= R_k(x + y - 1) \leq F_k(x + y - 1). \end{aligned}$$

Case 3. $x, y \geq k + 2$.

First assume that exactly one of $F_k(x) = J_k(x)$ and $F_k(y) = J_k(y)$ holds. By symmetry, assume that $F_k(x) > J_k(x)$ and $F_k(y) = J_k(y)$. Then there is an integer $b \in I_k$ such that $F_k(y) = J_k(y) = b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor$. By Claim 3.1 we have $F_k(x) \leq b(x - 1)$. Hence

$$F_k(x) + F_k(y) \leq b(x - 1) + b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor \leq J_k(x + y - 1) \leq F_k(x + y - 1).$$

Next suppose that $F_k(x) = J_k(x)$ and $F_k(y) = J_k(y)$. Let $a, b \in I_k$ be such that $F_k(x) = J_k(x) = a(x - a) + \left\lfloor \frac{a(k+1-a)}{2} \right\rfloor$ and $F_k(y) = J_k(y) = b(y - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor$. We may assume that $a \leq b$. Since $J_k(x + y - 1) \geq b(x + y - 1 - b) + \left\lfloor \frac{b(k+1-b)}{2} \right\rfloor$, we have $J_k(x + y - 1) - J_k(y) \geq b(x - 1)$. Because $a \leq b$ and $x \geq 1$, we obtain

$$\begin{aligned} F_k(x) + F_k(y) - J_k(x + y - 1) &\leq a(x - a) + \frac{a(k+1-a)}{2} - b(x - 1) \\ &= (a - b)(x - 1) + \frac{a(k+3-3a)}{2} \leq 0. \end{aligned}$$

Hence $F_k(x) + F_k(y) \leq J_k(x + y - 1) \leq F_k(x + y - 1)$.

Finally, assume that $F_k(x) > J_k(x)$ and $F_k(y) > J_k(y)$. Since $x + y \geq 2k + 4$, we have

$$\begin{aligned} F_k(x) + F_k(y) &= R_k(x) + R_k(y) \leq R_k(x + y) \leq \frac{(k+1)(x+y)}{2} \\ &\leq (k+1)(x+y-k-2) \leq J_k(x+y-1) \leq F_k(x+y-1). \end{aligned}$$

(ii) We first show that $F_k(n)$ is strictly increasing. For $1 \leq n \leq k$, $F_k(n+1) - F_k(n) = n \geq 1$. For $n = k+1$, $F_k(k+2) \geq R_k(k+2) > \binom{k+1}{2} = F_k(k+1)$. For $n \geq k+2$, $R_k(n+1) \geq R_k(n) + 1$ and $J_k(n+1) \geq J_k(n) + 1$, so $F_k(n+1) = \max\{R_k(n+1), J_k(n+1)\} > \max\{R_k(n), J_k(n)\} = F_k(n)$. Hence $F_k(n+1) > F_k(n)$ for all $n \geq 1$.

Since $x + y \leq n$, part (i) gives $F_k(x) + F_k(y) \leq F_k(x + y - 1) \leq F_k(x + y)$. ■

The next two lemmas provide the edge bounds used in the proof of Lemma 5.1.

Lemma 3.9. *Let $k \geq 2$ and $k + 2 \leq n \leq \lceil (5k+1)/2 \rceil$. Let $c < n$ be an integer. Then the following statements hold:*

- (i) *For every integer s with $2 \leq s \leq \lfloor \frac{c}{2} \rfloor - 1$, we have $s(n-c) + \left\lfloor \frac{(s-1)s + (c-s+1)(k+1)}{2} \right\rfloor \leq F_k(n)$.*
- (ii) *If $\lfloor c/2 \rfloor \geq 1$, then $(\lfloor \frac{c}{2} \rfloor - 1)(n-c) + \left\lfloor \frac{(\lfloor \frac{c}{2} \rfloor - 1) \min\{\lfloor \frac{c}{2} \rfloor, k+1\} + (c - \lfloor \frac{c}{2} \rfloor + 1)(k+1)}{2} \right\rfloor \leq F_k(n)$.*

Proof. We first consider $k = 2$. In this case $4 \leq n \leq 6$. Since $c < n$, we have $\lfloor c/2 \rfloor - 1 \leq 1$. Thus there is no integer s in the stated range in (i).

For (ii), since $k = 2$, we have $4 \leq n \leq 6$. We divide the proof into three cases.

Suppose that $n = 4$. We have $c \in \{2, 3\}$. The left-hand side takes the values 3 and 4, respectively. Since $F_2(4) = 6$, the inequality in (ii) holds.

Suppose that $n = 5$. We have $c \in \{2, 3, 4\}$. The left-hand side takes the values 3, 4, and 6, respectively. Since $F_2(5) = 7$, the inequality in (ii) holds.

Suppose that $n = 6$. We have $c \in \{2, 3, 4, 5\}$. The left-hand side takes the values 3, 4, 7, and 8, respectively. Since $F_2(6) = 9$, the inequality in (ii) holds.

Therefore, both (i) and (ii) hold when $k = 2$.

Assume that $k \geq 3$. Let $\varepsilon \in \{0, 1\}$ and let r be an integer satisfying $1 + \varepsilon \leq r \leq \lfloor c/2 \rfloor$. Set

$$L_\varepsilon(c, r) = (r-1)(n-c) + \left\lfloor \frac{(r-1-\varepsilon)(r-\varepsilon) + (c-r+1+\varepsilon)(k+1)}{2} \right\rfloor.$$

We now claim that $L_\varepsilon(c, r) \leq F_k(n)$.

We first compare $L_\varepsilon(c, r)$ with $\frac{(k+1)n}{2}$ using the following inequality:

$$2 \left(L_\varepsilon(c, r) - \frac{(k+1)n}{2} \right) \leq (n-c)(2r-k-3) - (r-1-\varepsilon)(k+1-r+\varepsilon). \quad (3.3)$$

If the right-hand side of (3.3) is nonpositive, then since $L_\varepsilon(c, r)$ is an integer, we have $L_\varepsilon(c, r) \leq R_k(n)$. We distinguish two cases.

Case 1. $J_k(n) \leq R_k(n)$.

Then $F_k(n) = R_k(n)$. By Lemma 3.5, we have $n \leq 2k + 2$. Since $2r \leq c < n$, it follows that $r \leq k$.

First, suppose that $2r - k - 3 \leq 0$. Since $n - c > 0$, $r - 1 - \varepsilon \geq 0$, and $k + 1 - r + \varepsilon \geq 0$, both terms on the right-hand side of (3.3) are nonpositive. Consequently, $L_\varepsilon(c, r) \leq R_k(n) = F_k(n)$.

Next, we assume that $2r - k - 3 > 0$. Then $r \geq \left\lfloor \frac{k+1}{2} \right\rfloor + 1$, and hence $r \in I_k$. Therefore

$$r(n-r) + \left\lfloor \frac{r(k+1-r)}{2} \right\rfloor \leq J_k(n) \leq R_k(n) \leq \frac{(k+1)n}{2}.$$

Since $\left\lfloor \frac{r(k+1-r)}{2} \right\rfloor \geq \frac{r(k+1-r)}{2} - \frac{1}{2}$, we obtain $r(n-r) + \frac{r(k+1-r)}{2} - \frac{1}{2} \leq \frac{(k+1)n}{2}$. Thus, $(2r-k-1)n \leq 2r^2 - r(k+1-r) + 1$. Hence

$$n \leq \frac{2r^2 - r(k+1-r) + 1}{2r-k-1}. \quad (3.4)$$

We claim that for $\varepsilon \in \{0, 1\}$, $(r-1-\varepsilon)(k+1-r+\varepsilon) > (2r-k-3)(n-2r)$. Indeed, from (3.4) and $2r-k-3 > 0$, we have $(2r-k-3)(n-2r) \leq \frac{(2r-k-3)(r(k+1-r)+1)}{2r-k-1}$. Thus, for $\varepsilon = 0$,

$$(r-1)(k+1-r) - (2r-k-3)(n-2r) \geq \frac{r(k-r) + (k+1-r)^2 + (k+1-r) + 2}{2r-k-1}, \quad (3.5)$$

and for $\varepsilon = 1$,

$$(r-2)(k+2-r) - (2r-k-3)(n-2r) \geq \frac{r(2r-k-4) + 2(k+1-r)^2 + 3(k+1-r) + 2}{2r-k-1}. \quad (3.6)$$

Since $2r-k-3 > 0$, the common denominator $2r-k-1$ in (3.5) and (3.6) is positive. Since $r \leq k$, the numerator in (3.5) satisfies

$$r(k-r) + (k+1-r)^2 + (k+1-r) + 2 = (k+3)(k-r) + 4 \geq 4.$$

Since $k \geq 3$, the numerator in (3.6) satisfies

$$\begin{aligned} & r(2r-k-4) + 2(k+1-r)^2 + 3(k+1-r) + 2 \\ &= 4r^2 - (5k+11)r + 2k^2 + 7k + 7 \\ &= 4 \left(r - \frac{5k+11}{8} \right)^2 + \frac{(k-1)(7k+9)}{16} > 0. \end{aligned}$$

Hence, the right-hand sides of (3.5) and (3.6) are positive. Then, for $\varepsilon \in \{0, 1\}$, $(r-1-\varepsilon)(k+1-r+\varepsilon) > (2r-k-3)(n-2r)$.

Since $c \geq 2r$, we have $n-c \leq n-2r$. It follows that $(2r-k-3)(n-c) \leq (r-1-\varepsilon)(k+1-r+\varepsilon)$. The right-hand side of (3.3) is nonpositive. Hence $L_\varepsilon(c, r) \leq R_k(n) = F_k(n)$.

Case 2. $J_k(n) > R_k(n)$.

If $k = 3$, then $5 \leq n \leq 8$ and $R_3(n) = 2n$, $J_3(n) = \max\{3n-8, 4n-16\} \leq 2n$. Thus, this case cannot occur when $k = 3$. Therefore, we may assume that $k \geq 4$. Then $F_k(n) = J_k(n)$, and Lemma 3.5 gives $n > \left(1 + \frac{\sqrt{3}}{2}\right)(k+1)$. If $2r-k-3 \leq 0$, then $r \leq k$, and the right-hand side of (3.3) is nonpositive. Hence $L_\varepsilon(c, r) \leq R_k(n) < J_k(n) = F_k(n)$. We may therefore assume that $2r-k-3 > 0$. In the following we prove that $L_\varepsilon(c, r) \leq J_k(n)$.

We first reduce the problem to the case $c = 2r$. Assume $c > 2r$. Since decreasing c by one decreases the numerator inside the floor of $L_\varepsilon(c, r)$ by $k+1$, we have $L_\varepsilon(c-1, r) - L_\varepsilon(c, r) \geq (r-1) - \left\lceil \frac{k+1}{2} \right\rceil$. Since $2r-k-3 > 0$, we have $r-1 \geq \left\lceil \frac{k+1}{2} \right\rceil$. Then $L_\varepsilon(c-1, r) \geq L_\varepsilon(c, r)$.

We can repeat this argument as c decreases to $2r$. Then $L_\varepsilon(c, r) \leq L_\varepsilon(2r, r)$. By dropping the floor, we have

$$L_\varepsilon(c, r) \leq L_\varepsilon(2r, r) \leq (r-1)(n-2r) + \frac{(r-1-\varepsilon)(r-\varepsilon) + (r+1+\varepsilon)(k+1)}{2}. \quad (3.7)$$

We next find a lower bound for $J_k(n)$. Note that

$$a(n-a) + \frac{a(k+1-a)}{2} = \frac{(2n+k+1)^2}{24} - \frac{3}{2} \left(a - \frac{2n+k+1}{6} \right)^2.$$

Since $\lfloor (1 + \frac{\sqrt{3}}{2})(k+1) \rfloor \leq n \leq \lceil \frac{(5k+1)}{2} \rceil$, every integer closest to $\frac{(2n+k+1)}{6}$ is in I_k . Choose an integer $a \in I_k$ nearest to $\frac{2n+k+1}{6}$. Then $|a - \frac{2n+k+1}{6}| \leq \frac{1}{2}$. It implies that $a(n-a) + \frac{a(k+1-a)}{2} \geq \frac{(2n+k+1)^2}{24} - \frac{3}{8}$. Since $a(n-a) + \frac{a(k+1-a)}{2}$ is an integer or a half-integer, we have $J_k(n) \geq \left\lfloor a(n-a) + \frac{a(k+1-a)}{2} \right\rfloor \geq a(n-a) + \frac{a(k+1-a)}{2} - \frac{1}{2}$. Therefore,

$$J_k(n) \geq \frac{(2n+k+1)^2}{24} - \frac{7}{8}. \quad (3.8)$$

First, suppose that $\varepsilon = 0$. By (3.7) and (3.8), $J_k(n) - L_0(c, r) \geq \frac{n}{2} - \frac{3(k+1)}{4} - \frac{5}{4}$. If $k \geq 6$, then $J_k(n) - L_0(c, r) > \frac{\sqrt{3}-1}{4}(k+1) - \frac{5}{4} \geq \frac{7\sqrt{3}-12}{4} > 0$. Thus, $L_0(c, r) < J_k(n)$ for $k \geq 6$. Assume that $k \in \{4, 5\}$. Since $2r - k - 3 > 0$ and $2r \leq c < n \leq \lceil \frac{5k+1}{2} \rceil$, we have $r \in \{k, k+1\}$.

If $r = k$, then $n \geq 10$ when $k = 4$ and $n \geq 12$ when $k = 5$. We have $L_0(2k, k) = (k-1)n - k^2 + 2k + \lfloor \frac{k+1}{2} \rfloor$. On the other hand, by the definition of $J_k(n)$, taking $a = k$ gives $J_k(n) \geq kn - k^2 + \lfloor \frac{k}{2} \rfloor$. Hence $J_k(n) - L_0(2k, k) \geq n - 2k + \lfloor \frac{k}{2} \rfloor - \lfloor \frac{k+1}{2} \rfloor \geq 1$.

If $r = k+1$, then $L_0(2k+2, k+1) = kn - k^2 + 1 \leq kn - k^2 + \lfloor \frac{k}{2} \rfloor \leq J_k(n)$. Therefore, $L_0(c, r) \leq J_k(n)$.

Now suppose that $\varepsilon = 1$. By (3.7) and (3.8), $J_k(n) - L_1(c, r) \geq \frac{5n}{6} - \frac{13(k+1)}{12} - \frac{23}{12}$. If $k \geq 4$, $J_k(n) - L_1(c, r, 1) > \frac{5\sqrt{3}-3}{12}(k+1) - \frac{23}{12} \geq \frac{25\sqrt{3}-38}{12} > 0$. Thus $L_1(c, r) < J_k(n)$.

Combining Cases 1 and 2, we obtain $L_\varepsilon(c, r) \leq F_k(n)$ for every $k \geq 3$ and all possible r and ε .

For (i), take $\varepsilon = 1$ and $r = s+1$. Since $2 \leq s \leq \lfloor c/2 \rfloor - 1$, $s(n-c) + \lfloor \frac{(s-1)s+(c-s+1)(k+1)}{2} \rfloor \leq F_k(n)$. This proves (i).

For (ii), take $\varepsilon = 0$ and $r = \lfloor c/2 \rfloor$. Since $r \geq 1$, $(r-1)(n-c) + \lfloor \frac{(r-1)r+(c-r+1)(k+1)}{2} \rfloor \leq F_k(n)$. If $r \leq k+1$, then $\min\{r, k+1\} = r$. Then the inequality in (ii) follows immediately. If $r > k+1$, then $(r-1)(k+1) \leq (r-1)r$. Hence $\lfloor \frac{(r-1)(k+1)+(c-r+1)(k+1)}{2} \rfloor \leq \lfloor \frac{(r-1)r+(c-r+1)(k+1)}{2} \rfloor$. This proves (ii). $\blacksquare$

Lemma 3.10. *Let $2 \leq k \leq 6$, let $k+2 \leq n \leq \lceil (5k+1)/2 \rceil$, and let c be an integer satisfying $2\lfloor (k+1)/2 \rfloor + 2 \leq c < n$ and $c \leq 9$. Then*

$$\left\lfloor \frac{c}{2} \right\rfloor (n-c) + \left\lfloor \frac{(\lfloor \frac{c}{2} \rfloor - 1)\lfloor \frac{c}{2} \rfloor + (c - \lfloor \frac{c}{2} \rfloor + 1)(k+1)}{2} \right\rfloor \leq F_k(n), \quad (3.9)$$

except when $(k, n, c) = (3, 8, 6)$. If $(k, n, c) = (3, 8, 6)$, the left-hand side of (3.9) is $17 = F_3(8) + 1$.

Proof. We first list all possible values of (k, c) in Table 1:

k	c	left-hand side of (3.9)
2	4, 5	$2n - 3$
3	6, 7	$3n - (c + 1)$
4	6, 7	$3n - (c - 1)$
4	8, 9	$4n - (c + 6)$
5	8, 9	$4n - (c + 3)$
6	8, 9	$4n - 9$

Table 1: $2 \leq k \leq 6$

Suppose that $k = 2$. For $n = 5$, $F_2(5) = \max\{\lfloor 15/2 \rfloor, 2 \cdot 5 - 3\} = 7$. For $n = 6$, $F_2(6) = \max\{9, 9\} = 9$. Hence $2n - 3 = F_2(n)$.

Suppose that $k = 3$. The possible pairs (n, c) are $(7, 6)$, $(8, 6)$, $(8, 7)$. The left-hand side gives 14, 17, 16, while $F_3(7) = 14$ and $F_3(8) = 16$. The only exception is $(k, n, c) = (3, 8, 6)$.

Suppose that $k = 4$. If $c = 6$, the left-hand side of (3.9) takes value $3n - 5$. For $n \leq 10$, we have $3n - 5 \leq \lfloor 5n/2 \rfloor = R_4(n) \leq F_4(n)$. For $n = 11$, we have $3 \cdot 11 - 5 = 28 < 30 = F_4(11)$. If $c = 7$, then $3n - 6 \leq \lfloor 5n/2 \rfloor \leq F_4(n)$. If $c \in \{8, 9\}$, the left-hand side of (3.9) takes values $4n - 14$ or $4n - 15$. Hence (3.9) holds because $J_4(n) = 4n - 14 \leq F_4(n)$.

Suppose that $k = 5$. For $n \leq 11$, we have $4n - 11 \leq 3n \leq F_5(n)$. For $n \geq 12$, both $4n - 11$ and $4n - 12$ are at most $J_5(n) = 5n - 23 \leq F_5(n)$.

Suppose that $k = 6$. Since $n \leq 16$, we obtain $4n - 9 \leq \lfloor \frac{7n}{2} \rfloor = R_6(n) \leq F_6(n)$.

This completes the proof. ■

4 Tools: closure, edge-switching, and stability

We collect the closure, edge-switching, and circumference results used in the proof of Theorem 1.3. These results will be used to bound the number of edges inside C and the number of edges with at least one endpoint outside C .

4.1 Three families of graphs

We use the graph $W_{n,s,t}$ and the exceptional families $\mathcal{X}_{n,c}$ and $\mathcal{Y}_{n,c}$ from the stability results in [13, 14, 18]. The families $\mathcal{X}_{n,c}$ and $\mathcal{Y}_{n,c}$ are defined for odd c .

- For integers n, s, t with $1 \leq s \leq \lfloor (t+1)/2 \rfloor$ and $t - s + 1 \leq n$, define $W_{n,s,t}$ as follows. Let S, T, U be pairwise disjoint sets with $|S| = s$, $|T| = t - 2s + 1$, and $|U| = n - t + s - 1$. Make $S \cup T$ a clique, make U independent, and join each vertex of U to the vertices of S . The case $U = \emptyset$ is allowed. Equivalently, $W_{n,s,t}$ is obtained from a clique K_{t-s+1} by adding $n - (t - s + 1)$ vertices whose common neighborhood is a fixed s -set in the clique. Thus $e(W_{n,s,t}) = \binom{t-s+1}{2} + s(n - t + s - 1)$.
- A graph G belongs to $\mathcal{X}_{n,c}$ if it has order n and a partition $A \cup B \cup X$ such that $G[A]$ induces a clique $K_{\lfloor c/2 \rfloor}$, both B and X are independent, (A, B) is complete bipartite, and there exist $a \in A$ and $b \in B$ such that $N_G(x) = \{a, b\}$ for every $x \in X$.
- A graph G belongs to $\mathcal{Y}_{n,c}$ if it has order n and a partition $A \cup B \cup Y$ such that $G[A]$ induces a clique $K_{\lfloor c/2 \rfloor}$, B is independent, (A, B) is complete bipartite, and $G[Y]$ is a nontrivial star forest (with at least two components, each containing an edge). There are vertices $a, b \in A$ such that every component R of $G[Y]$ satisfies $N_G(V(R)) = \{a, b\}$. If $|V(R)| \geq 3$, then all leaves of R have degree 2 in G and have a common neighbor in $\{a, b\}$.

4.2 Classical and stability tools

Theorem 4.1 (Dirac [9]). *If G is a 2-connected graph with minimum degree $\delta(G)$, then $c(G) \geq \min\{|V(G)|, 2\delta(G)\}$. In particular, if G is non-Hamiltonian, then $c(G) \geq 2\delta(G)$.*

Theorem 4.2 (Ma–Ning [18]). *Let G be a 2-connected graph on n vertices, and let C be a longest cycle of length c , where $10 \leq c \leq n - 1$. If*

$$e(G - C) + e_G(V(G) \setminus V(C), V(C)) > (\lfloor c/2 \rfloor - 1)(n - c),$$

then either $G \subseteq W_{n, \lfloor c/2 \rfloor, c}$, or c is odd and G is a subgraph of a member of $\mathcal{X}_{n,c} \cup \mathcal{Y}_{n,c}$.

Lemma 4.3. *Let $c < n$ be odd and let $G \in \mathcal{X}_{n,c} \cup \mathcal{Y}_{n,c}$. Then either $G \subseteq W_{n, \lfloor c/2 \rfloor, c}$, or G has a vertex of degree at most 3.*

Proof. First, suppose that $G \in \mathcal{X}_{n,c}$, with partition $V(G) = A \cup B \cup X_0$. If $X_0 \neq \emptyset$, then every vertex of X_0 has neighborhood exactly $\{a, b\}$ and hence degree 2.

Thus assume $X_0 = \emptyset$. Then $V(G) = A \cup B$ such that $G[A]$ induces a clique $K_{\lfloor c/2 \rfloor}$, B is independent, (A, B) is complete bipartite. Since c is odd and $n > c$, we have $|B| = n - \lfloor c/2 \rfloor \geq \lfloor c/2 \rfloor + 2 \geq 2$. Choose distinct $b_1, b_2 \in B$. In the partition of $W_{n, \lfloor c/2 \rfloor, c}$, take $S = A$, $T = \{b_1, b_2\}$, $U = B \setminus \{b_1, b_2\}$. Indeed, $|T| = c - 2 \lfloor c/2 \rfloor + 1 = 2$. Then every edge of G is an edge of this copy of $W_{n, \lfloor c/2 \rfloor, c}$. Hence $G \subseteq W_{n, \lfloor c/2 \rfloor, c}$.

Now suppose that $G \in \mathcal{Y}_{n,c}$, with partition $V(G) = A \cup B \cup Y_0$. If some star component of $G[Y_0]$ has order at least 3, then one of its leaves has degree 2 in G . Otherwise every star component of $G[Y_0]$ has order 2. For any such component uv , the condition $N_G(\{u, v\}) = \{a, b\}$ gives $d_G(u), d_G(v) \leq 3$. Thus G has a vertex of degree at most 3. ■

4.3 Some results on cycles

The following lemma is a slight variant of a lemma of Ma and Ning [18, Lemma 2.11]. It will be used to obtain the edge bound.

Lemma 4.4 (Variant of Ma–Ning [18, Lemma 2.11]). *Let G be a 2-connected graph and let C be a locally maximal cycle of length $c < |V(G)|$. Set $\hat{G} = \text{cl}_C(G)$ and $\hat{H} = \hat{G}[V(C)]$. Then one of the following holds.*

- (i) *There exist an integer s and a set $S \subseteq V(C)$ such that $2 \leq s \leq \lfloor c/2 \rfloor - 1$, $|S| = s - 1$, $d_{\hat{H}}(u) \leq s$ for every $u \in S$, and $\hat{H} - S$ is a clique.*
- (ii) *There exists a set $R \subseteq V(C)$ such that $|R| = \lfloor c/2 \rfloor - 1$ and $d_{\hat{H}}(u) \leq \lfloor c/2 \rfloor$ for every $u \in R$.*

To prove the lemma, we need some results of Ma and Ning [18].

Lemma 4.5 (Ma–Ning [18, Lemma 2.7]). *Let G be a graph and let C be a locally maximal cycle of G . Then C is locally maximal in $\text{cl}_C(G)$.*

Lemma 4.6 (Ma–Ning [18, Lemma 2.8]). *Let G be a 2-connected graph on n vertices and let C be a locally maximal cycle of length $c < n$. Then $\text{cl}_C(G)[V(C)]$ is non-Hamiltonian-connected.*

Lemma 4.7 (Ma–Ning [18, Lemma 2.10]). *Let G be a non-Hamiltonian-connected graph on n vertices with minimum degree at least 2. Then there exists an integer s with $2 \leq s \leq \lfloor n/2 \rfloor$ such that G has at least $s - 1$ vertices of degree at most s .*

We now prove Lemma 4.4.

Proof of Lemma 4.4. By Lemma 4.6, $\hat{H}$ is non-Hamiltonian-connected. Since $C \subseteq \hat{H}$, we have $\delta(\hat{H}) \geq 2$. Thus Lemma 4.7 applies. Then there exists an integer s with $2 \leq s \leq \lfloor c/2 \rfloor$ such that $\hat{H}$ has at least $s - 1$ vertices of degree at most s . Take such an s to be maximal.

If $s = \lfloor c/2 \rfloor$, then case (ii) holds by taking any $\lfloor c/2 \rfloor - 1$ of these vertices. Thus we may assume $s \leq \lfloor c/2 \rfloor - 1$. Let $S_0 = \{v \in V(C) : d_{\hat{H}}(v) \leq s\}$. We show that either $\hat{H} - S_0$ is a clique, which gives (i), or (ii) holds. By the choice of s , we have $|S_0| \geq s - 1$. If $|S_0| \geq s$, then there are at least s vertices of degree at most s . Since $s + 1 \leq \lfloor c/2 \rfloor$, this contradicts the maximality of s . Hence $|S_0| = s - 1$.

If $\hat{H} - S_0$ is a clique, then alternative (i) holds. Thus, we may assume that $\hat{H} - S_0$ is not a clique. We show that alternative (ii) holds under this assumption. Let $X = \{x \in V(\hat{H}) \setminus S_0 : x \text{ has a non-neighbor in } \hat{H} - S_0\}$. Choose two nonadjacent vertices $u, v \in X$ such that $d_{\hat{H}}(u)$ is maximum among all vertices in X . Set $S' = V(C) \setminus (N_{\hat{H}}(u) \cup \{u\})$ and $s' = |S'| + 1 = c - d_{\hat{H}}(u)$.

For every $w \in S'$, w and u are nonadjacent. Since $\hat{H}$ is $(c + 1)$ -closed, $d_{\hat{H}}(w) + d_{\hat{H}}(u) \leq c$. Hence $d_{\hat{H}}(w) \leq s'$. Since $v \in S' \setminus S_0$, we have $s < d_{\hat{H}}(v) \leq s'$. If $s' \leq \lfloor c/2 \rfloor$, then $\hat{H}$ has at least $s' - 1$ vertices of degree at most s' , contradicting the maximality of s . Therefore $s' \geq \lfloor c/2 \rfloor + 1$. It follows that $|S'| \geq \lfloor c/2 \rfloor$. Since $d_{\hat{H}}(u) = c - s'$, we have $d_{\hat{H}}(u) \leq c - \lfloor c/2 \rfloor - 1 \leq \lfloor c/2 \rfloor$.

If $w \in S' \cap S_0$, then $d_{\widehat{H}}(w) \leq s \leq \lfloor c/2 \rfloor$. If $w \in S' \setminus S_0$, then the choice of u implies $d_{\widehat{H}}(w) \leq d_{\widehat{H}}(u) \leq \lfloor c/2 \rfloor$. Hence, every vertex of S' has degree at most $\lfloor c/2 \rfloor$. Therefore, any $\lfloor c/2 \rfloor - 1$ vertices of S' form the desired set R in alternative (ii). ■

4.4 The number of edges with at least one endpoint outside C

The next lemma bounds the number of edges with at least one endpoint outside a locally maximal cycle. For a longest cycle C of length c in an n -vertex graph, Bondy [3] proved that (a) the number of edges with at most one endpoint in C is at most $\frac{c}{2}(n - c)$; (b) if the graph is 2-connected, the sharper bound $\lfloor \frac{c}{2} \rfloor (n - c)$ holds. In the following lemma, we extend (b) from longest cycles to locally maximal cycles.

Lemma 4.8. *Let G be a 2-connected graph of order n , and let C be a locally maximal cycle of length $c < n$.*

- (i) *Let $\widehat{G} = \text{cl}_C(G)$. If there exists a set $S \subseteq V(C)$ with $|S| = s - 1$, $2 \leq s \leq \lfloor \frac{c}{2} \rfloor - 1$, such that $\widehat{G}[V(C)] - S$ is a clique, then $e(G) \leq e(G[V(C)]) + s(n - c)$.*
- (ii) *If $c \geq 4$, then $e(G - C) + e_G(V(G) \setminus V(C), V(C)) \leq \lfloor \frac{c}{2} \rfloor (n - c)$.*

To prove this, we will need some tools developed by Ma and Ning [18] and Fan, Lv, and Wang [12]. Let G be a graph, let C be a cycle of G , and let Q be a component of $G - C$. A set $T = \{x_1, \dots, x_t\} \subseteq V(C)$ with $t \geq 2$, listed in cyclic order on C , is a *strong attachment* of Q to C if, for every $i \in \{1, \dots, t\}$, where indices are taken modulo t , there exist vertices $u_i, v_i \in V(Q)$ such that $x_i u_i, x_{i+1} v_i \in E(G)$ and $x_i u_i, x_{i+1} v_i$ are independent edges.

Lemma 4.9 (Ma–Ning [18, Lemma 2.5(i)]). *Let G be a 2-connected graph, let C be a locally maximal cycle of length c in G , and let Q be a component of $G - C$. Suppose T is a strong attachment of Q to C , $|T| = t$, and for every distinct $x, x' \in T$, a longest (x, Q, x') -path has length at least $d \geq 2$. Then $\omega(G[V(C)]) \leq c - (d - 1)(t - 1)$.*

Let G be a graph and let $xy \in E(G)$. Suppose $A \subseteq N_G(y) \setminus (N_G(x) \cup \{x\})$. We define the edge switch from y to x over A as follows. Let $G[y \rightarrow x; A]$ be obtained from G by replacing yz with xz for every $z \in A$. Thus, the operation transfers the adjacencies of y with the vertices in A to x , while keeping the number of edges unchanged.

Lemma 4.10 (Fan–Lv–Wang [12, Lemma 2.4]). *Let G be a 2-connected graph, let C be a locally maximal cycle in G , and let Q be a component of $G - C$. Then one of the following holds.*

- (i) *$N_Q(x) = V(Q)$ for every $x \in N_C(Q)$.*
- (ii) *There exist $x \in N_C(Q)$, $y \in N_Q(x)$, and a nonempty set $A \subseteq N_Q(y) \setminus N_Q(x)$ such that, with $G_0 = G[y \rightarrow x; A]$, one of the following holds:*
 - (a) *G_0 is 2-connected. In this case, set $G' = G_0$.*
 - (b) *G_0 is not 2-connected, and there exists $x' \in N_C(Q) \setminus \{x\}$ such that $G_0 + yx'$ is 2-connected. In this case, set $G' = G_0 + yx'$.*

If alternative (ii) occurs, then C remains locally maximal in G' , $G'[V(C)] = G[V(C)]$, and $e(G') \geq e(G)$.

Lemma 4.11 (Erdős–Gallai [11]). *Let G be a graph on n vertices whose longest path has length at most ℓ . Then $e(G) \leq \ell n/2$.*

We now prove Lemma 4.8. The framework used in the proof is adapted from the proof of Lemma 4.3 in Ma and Ning [18]. More precisely, following their argument, we repeatedly apply the edge-switching lemma of Fan, Lv, and Wang [12, Lemma 2.4] to obtain a terminal graph in which, for every component Q outside C , each vertex of $N_C(Q)$ is adjacent to every vertex of Q . We then combine the Erdős–Gallai path bound with the local maximality of C .

Proof of Lemma 4.8. We first prove the following claim about repeated edge switching:

Claim 4.1. *Let F be a 2-connected graph in which C is a locally maximal cycle of length $c < n$. We apply Lemma 4.10(ii) repeatedly, starting from F , to obtain a graph $\tilde{F}$. This graph has the following properties:*

- (a) $e(\tilde{F}) \geq e(F)$, $\tilde{F}[V(C)] = F[V(C)]$, and C is locally maximal in $\tilde{F}$;
- (b) every component Q of $\tilde{F} - C$ satisfies $N_Q(x) = V(Q)$ for every $x \in N_C(Q)$;
- (c) let $Q_1, \dots, Q_m$ be the components of $\tilde{F} - C$, let $q_i = |N_C(Q_i)|$ and let ℓ_i be the maximum length of a path in Q_i . Choose $Q = Q_j$ so that $\ell_j + 2q_j = \max_i(\ell_i + 2q_i)$. Let $\ell = \ell_j$ and $q = q_j$. Then we have $e(\tilde{F}) \leq e(\tilde{F}[V(C)]) + \frac{\ell+2q}{2}(n-c)$.

Proof. Let $F_0 = F$, and for $j \geq 1$, let F_j be the graph obtained from F_{j-1} by one application of Lemma 4.10(ii). If every component of $F_j - C$ satisfies alternative (i) of Lemma 4.10, then we stop. Otherwise, choose a component Q of $F_j - C$ for which alternative (i) fails. By Lemma 4.10, alternative (ii) applies. Denote the resulting graph by F_{j+1} . Since the switched set is nonempty, $e(F_{j+1} - C) < e(F_j - C)$. Hence the process terminates. Let $\tilde{F}$ be the terminal graph. Then $e(\tilde{F}) \geq e(F)$, $\tilde{F}[V(C)] = F[V(C)]$, C is locally maximal in $\tilde{F}$, and every component Q of $\tilde{F} - C$ satisfies $N_Q(x) = V(Q)$ for every $x \in N_C(Q)$. Thus (a) and (b) hold.

For each component Q_i of $\tilde{F} - C$, let $q_i = |N_C(Q_i)|$ and let ℓ_i be the maximum length of a path in Q_i . By (b), every vertex of $N_C(Q_i)$ is adjacent to all vertices of Q_i . Hence $e_{\tilde{F}}(V(Q_i), V(C)) = q_i|Q_i|$. Since the subgraph Q_i has no path longer than ℓ_i , Lemma 4.11 gives $e(\tilde{F}[Q_i]) \leq \frac{\ell_i|Q_i|}{2}$. Thus, $e(\tilde{F}[Q_i]) + e_{\tilde{F}}(V(Q_i), V(C)) \leq \left(\frac{\ell_i}{2} + q_i\right)|Q_i| = \frac{\ell_i + 2q_i}{2}|Q_i|$. Now choose a component Q_j maximizing $\ell_j + 2q_j$, and let $\ell = \ell_j$, $q = q_j$. Summing over all components, we obtain

$$\sum_i (e(\tilde{F}[Q_i]) + e_{\tilde{F}}(V(Q_i), V(C))) \leq \sum_i \frac{\ell_i + 2q_i}{2}|Q_i| \leq \frac{\ell + 2q}{2} \sum_i |Q_i| = \frac{\ell + 2q}{2}(n - c).$$

Since the graph $\tilde{F}$ consists of the subgraph inside C , the components outside C , and the edges joining them to C , we have

$$e(\tilde{F}) \leq e(\tilde{F}[V(C)]) + \frac{\ell + 2q}{2}(n - c).$$

Thus (c) holds. □

(i) Let $\hat{G} = \text{cl}_C(G)$. Since adding edges preserves 2-connectivity, $\hat{G}$ is 2-connected. By Lemma 4.5, C is locally maximal in $\hat{G}$. Then $e(G) - e(G[V(C)]) = e(\hat{G}) - e(\hat{G}[V(C)])$. Apply Claim 4.1 with $F = \hat{G}$, and let $\tilde{F}$ be the resulting terminal graph. Then, $e(\hat{G}) \leq e(\tilde{F})$ and $\tilde{F}[V(C)] = \hat{G}[V(C)]$. Let Q be the component given in Claim 4.1, let $q = |N_C(Q)|$, and let ℓ be the maximum length of a path in Q . Thus, $e(\hat{G}) - e(\hat{G}[V(C)]) \leq e(\tilde{F}) - e(\tilde{F}[V(C)]) \leq \frac{\ell+2q}{2}(n-c)$. It remains to show $\ell + 2q \leq 2s$.

First, suppose that $\ell = 0$. Then Q consists of a single vertex, say $Q = \{z\}$. Fix an orientation of C and set $U = \{x^+ : x \in N_C(Q)\}$. Since the successor map on C is injective, $|U| = q$. We claim that U is an independent set in $\tilde{F}[V(C)]$.

Suppose to the contrary that $x^+, y^+ \in U$ are distinct and $x^+y^+ \in E(\tilde{F})$ for distinct $x, y \in N_C(Q)$. If $x^+ = y$ or $y^+ = x$, then replacing xy of C by xzy gives a cycle of length $c + 1$, contradicting the local maximality of C . Otherwise, deleting xx^+ and yy^+ from C and adding xz , zy , x^+y^+ gives a cycle of length $c + 1$, again contradicting the local maximality of C . This proves that U is independent.

By hypothesis, $\hat{G}[V(C)] - S$ is a clique and $|S| = s - 1$. Consequently, every independent set in $\tilde{F}[V(C)]$ contains at most one vertex outside S . Hence every independent set has size at most $|S| + 1 = s$. Therefore $q = |U| \leq s$. Thus, $\ell + 2q = 2q \leq 2s$.

Now suppose that $\ell \geq 1$. Suppose to the contrary that $\ell + 2q \geq 2s$. Let $P = uPv$ be a longest path of length ℓ . Since $\tilde{F}$ is 2-connected, Q has at least two distinct neighbors on C . Hence

$q \geq 2$. By Claim 4.1(b), every vertex of $N_C(Q)$ is adjacent to every vertex of Q . Thus, for any two distinct vertices $x, x' \in N_C(Q)$, $xuPvx'$ is an (x, Q, x') -path of length $\ell + 2$. Since $x \neq x'$ and $u \neq v$, xu and $x'v$ are vertex-disjoint. Hence $N_C(Q)$ is a strong attachment of Q to C . By Lemma 4.9, $\omega(\tilde{F}[V(C)]) \leq c - (\ell + 1)(q - 1)$. Since $(\ell + 1)(q - 1) - (\ell + 2q - 3) = (\ell - 1)(q - 2) \geq 0$, we have $(\ell + 1)(q - 1) \geq \ell + 2q - 3$. If $s = 2$, then $(\ell + 1)(q - 1) \geq 2 = s$. If $s \geq 3$, then $(\ell + 1)(q - 1) \geq 2s - 3 \geq s$. It follows that $\omega(\tilde{F}[V(C)]) \leq c - s$. By hypothesis, $\tilde{G}[V(C)] - S$ is a clique of order $c - s + 1$. Thus, $\omega(\tilde{F}[V(C)]) \geq c - s + 1$, contradicting $\omega(\tilde{F}[V(C)]) \leq c - s$. Hence $\ell + 2q \leq 2s$. This proves (i).

(ii) Apply Claim 4.1 with $F = G$, and let $\tilde{G}$ be the resulting terminal graph. Then $e(\tilde{G}) \geq e(G)$, $\tilde{G}[V(C)] = G[V(C)]$. It follows that $e(G - C) + e_G(G - C, C) \leq e(\tilde{G} - C) + e_{\tilde{G}}(\tilde{G} - C, C)$.

Let Q be the component given in Claim 4.1, let $q = |N_C(Q)|$, and let ℓ be the maximum length of a path in Q . Since $\tilde{G}$ is 2-connected, $q \geq 2$. Set $N_C(Q) = \{x_1, \dots, x_q\}$ in cyclic order on C , with indices taken modulo q . Let $C_i = x_i C x_{i+1}$ and let P be a longest path in Q . By Claim 4.1(b), every vertex of $N_C(Q)$ is adjacent to every vertex of Q . If $\ell = 0$, the unique vertex of P gives an (x_i, Q, x_{i+1}) -path of length 2. If $\ell \geq 1$, adding x_i and x_{i+1} to the two ends of P gives such a path of length $\ell + 2$. Thus, in either case, there is an (x_i, Q, x_{i+1}) -path of length $\ell + 2$. If $|E(C_i)| < \ell + 2$ for some i , replacing C_i by the (x_i, Q, x_{i+1}) -path of length $\ell + 2$ will give a cycle longer than C . The new cycle uses exactly two edges between C and $\tilde{G} - C$, contradicting the local maximality of C . Therefore $|E(C_i)| \geq \ell + 2$ for every i . Summing over i yields $(\ell + 2)q \leq c$. Thus, $\ell + 2q \leq 2 \lfloor \frac{c}{2} \rfloor$. By Claim 4.1(c),

$$e(\tilde{G} - C) + e_{\tilde{G}}(V(\tilde{G}) \setminus V(C), V(C)) = e(\tilde{G}) - e(\tilde{G}[V(C)]) \leq \frac{\ell + 2q}{2}(n - c) \leq \left\lfloor \frac{c}{2} \right\rfloor (n - c).$$

This proves (ii). ■

5 Proof of Theorem 1.3

It suffices to prove the following two lemmas.

Lemma 5.1. *Let $k \geq 2$ and $k + 2 \leq n \leq \lceil (5k + 1)/2 \rceil$. Then every n -vertex graph containing no $(k + 2)$ -path-fan has at most $F_k(n)$ edges.*

Lemma 5.2. *Let $k \geq 2$ and $n \geq \lceil (5k + 1)/2 \rceil$. Then every n -vertex graph containing no $(k + 2)$ -path-fan has at most $(k + 1)(n - k - 1)$ edges.*

We first prove Lemma 5.1. To prove it, we prove two helpful lemmas, i.e., Lemmas 5.3 and 5.4.

Lemma 5.3. *Let $k \geq 1$, let G be a graph containing no $(k + 2)$ -path-fan, and let C be any cycle of G . Then $d_{G[V(C)]}(v) \leq k + 1$ for each $v \in V(C)$.*

Proof. If $d_{G[V(C)]}(v) \geq k + 2$ for some $v \in V(C)$, then $C - v$ is a path in $G - v$ containing at least $k + 2$ neighbors of v . Thus v and $C - v$ form a $(k + 2)$ -path-fan, a contradiction. ■

Lemma 5.4. *Fix $k \geq 2$. Suppose that Lemma 5.1 is false. Choose a counterexample G with $n := |V(G)|$ minimum and, subject to this, with $e(G)$ minimum. Then the following hold:*

- (1) $e(G) = F_k(n) + 1$;
- (2) G is non-Hamiltonian;
- (3) G is 2-connected;
- (4) $\delta(G) \geq \lfloor (k + 1)/2 \rfloor + 1$;
- (5) If C is a longest cycle of G with length c , then $2\lfloor (k + 1)/2 \rfloor + 2 \leq c < n$.

Proof. (1) Since deleting edges cannot create a $(k+2)$ -path-fan, the choice of G gives $e(G) = F_k(n) + 1$.

(2) Since $F_k(n) \geq R_k(n)$, we have $e(G) > (k+1)n/2$. Thus some vertex x has degree at least $k+2$. If C is a Hamilton cycle, then $C - x$ and x form a $(k+2)$ -path-fan. Hence G is non-Hamiltonian.

For (3) and (4), we use the following two properties.

Claim 5.1. (i) For every integer $m \geq 1$ and every $a \in I_k$ such that $J_k(m) = a(m-a) + \lfloor \frac{a(k+1-a)}{2} \rfloor$, we have $J_k(m+1) \geq J_k(m) + a$.
(ii) If $F_k(n) = R_k(n)$, then $F_k(m) = R_k(m)$ for every $k+2 \leq m < n$.

Proof. (i) By the definition of $J_k(m+1)$, $J_k(m+1) \geq a(m+1-a) + \lfloor \frac{a(k+1-a)}{2} \rfloor = J_k(m) + a$.

(ii) For $k+2 \leq m < n$, let $a \in I_k$ such that $J_k(m) = a(m-a) + \lfloor \frac{a(k+1-a)}{2} \rfloor$. By (i), we have $J_k(n) \geq a(n-a) + \lfloor \frac{a(k+1-a)}{2} \rfloor = J_k(m) + (n-m)a$. Since $F_k(n) = R_k(n)$, we have $J_k(n) \leq R_k(n)$. Since $a \geq \lfloor (k+1)/2 \rfloor + 1 > (k+1)/2$, we have $(n-m)a + \lfloor (k+1)m/2 \rfloor \geq \lfloor (k+1)n/2 \rfloor = R_k(n)$. Then $J_k(m) - R_k(m) \leq J_k(n) - (n-m)a - R_k(m) = J_k(n) - ((n-m)a + R_k(m)) \leq J_k(n) - R_k(n) \leq 0$. Hence $J_k(m) \leq R_k(m)$, which implies $F_k(m) = R_k(m)$. This proves (ii). $\square$

(3) First, suppose that G is disconnected, and let $Q_1, \dots, Q_t$ be its components, where $q_i := |V(Q_i)|$. If $q_i \leq k+1$, then $e(Q_i) \leq \binom{q_i}{2} = F_k(q_i)$. If $q_i \geq k+2$, then the minimality of n gives $e(Q_i) \leq F_k(q_i)$. Thus, $e(Q_i) \leq F_k(q_i)$ for every i . If $J_k(n) > R_k(n)$, repeated application of Lemma 3.8(ii) gives $e(G) \leq \sum_i F_k(q_i) \leq F_k(n)$, contradicting (1). If $J_k(n) \leq R_k(n)$, Claim 5.1(ii) and the definition of F_k give $e(G) \leq \sum_i F_k(q_i) \leq \sum_i \lfloor \frac{(k+1)q_i}{2} \rfloor \leq R_k(n) = F_k(n)$, a contradiction. Hence G is connected.

Suppose that z is a cut-vertex. Let $Q_1, \dots, Q_t$ be the components of $G - z$, and set $G_i := G[V(Q_i) \cup \{z\}]$ and $n_i := |V(G_i)|$. If $n_i \leq k+1$, then $e(G_i) \leq \binom{n_i}{2} = F_k(n_i)$. If $n_i \geq k+2$, then the minimality of n gives $e(G_i) \leq F_k(n_i)$. Thus, $e(G) \leq \sum_i e(G_i) \leq \sum_i F_k(n_i)$. We consider the cases $J_k(n) > R_k(n)$ and $J_k(n) \leq R_k(n)$. If $J_k(n) > R_k(n)$, then by Lemma 3.8(i), $e(G) \leq \sum_i F_k(n_i) \leq F_k(\sum_i n_i - (t-1)) = F_k(n) = J_k(n)$, contradicting (1). If $R_k(n) \geq J_k(n)$, by Lemma 3.5(ii), $n \leq 2k+2$. By Claim 5.1(ii), if $n_i \geq k+2$, $F_k(n_i) = R_k(n_i) \leq \lfloor (k+1)n_i/2 \rfloor$. Define $I = \{i : n_i \geq k+2\}$. If $|I| \geq 2$, then $n = 1 + \sum_i (n_i - 1) \geq 1 + 2(k+1) = 2k+3$, contradicting $n \leq 2k+2$. Hence $|I| \leq 1$. If $I = \emptyset$, then $e(G) \leq \sum_i \binom{n_i}{2} \leq \sum_i \lfloor \frac{(k+1)(n_i-1)}{2} \rfloor \leq \lfloor \frac{(k+1)(n-1)}{2} \rfloor < R_k(n)$. If $I = \{i_0\}$, then $e(G) \leq R_k(n_{i_0}) + \sum_{i \neq i_0} \binom{n_i}{2} \leq \lfloor \frac{(k+1)n_{i_0}}{2} \rfloor + \sum_{i \neq i_0} \lfloor \frac{(k+1)(n_i-1)}{2} \rfloor \leq R_k(n)$, contradicting (1). Hence G has no cut-vertex. This proves (3).

(4) Let $v \in V(G)$. If $n-1 \leq k+1$, then $e(G-v) \leq F_k(n-1) \leq R_k(n-1)$. Hence $d_G(v) \geq R_k(n) + 1 - R_k(n-1) \geq \lfloor (k+1)/2 \rfloor + 1$. We may therefore assume that $n-1 \geq k+2$. By the minimality of n , $d_G(v) = e(G) - e(G-v) \geq F_k(n) + 1 - F_k(n-1)$. If $J_k(n) \leq R_k(n)$, then Claim 5.1(ii) gives $F_k(n-1) = R_k(n-1)$. Thus, $d_G(v) \geq R_k(n) + 1 - R_k(n-1) \geq \lfloor (k+1)/2 \rfloor + 1$. Suppose that $J_k(n) > R_k(n)$. If $F_k(n-1) = J_k(n-1)$, then Claim 5.1(i) gives $F_k(n) - F_k(n-1) \geq \lfloor (k+1)/2 \rfloor + 1$. If $F_k(n-1) = R_k(n-1)$, then $F_k(n) \geq R_k(n) + 1$. Therefore $d_G(v) \geq \lfloor (k+1)/2 \rfloor + 1$. This proves (4).

(5) Let C be a longest cycle of G and let $c = |E(C)|$. By (2) and (3), the graph G is non-Hamiltonian and 2-connected. Hence Theorem 4.1 gives $n > c \geq 2\delta(G) \geq 2 \lfloor (k+1)/2 \rfloor + 2$. This completes the proof. $\blacksquare$

Now we are in a position to prove Lemma 5.1.

Proof of Lemma 5.1. Suppose that Lemma 5.1 is false, and let G be the minimal counterexample given in Lemma 5.4. Let C be a longest cycle of G . Set $H = G[V(C)]$ and $c = |E(C)|$. Let $\hat{G} = \text{cl}_C(G)$, $\hat{H} = \hat{G}[V(C)]$. Since G is 2-connected and non-Hamiltonian, Lemma 4.4 applies. If Lemma 4.4 (i) holds, there exist an integer s and a set $S \subseteq V(C)$ such

that $2 \leq s \leq \lfloor \frac{c}{2} \rfloor - 1$, $|S| = s - 1$, $d_{\widehat{H}}(u) \leq s$ for every $u \in S$, and $\widehat{H} - S$ is a clique. By Lemma 4.8(i), $e(G) \leq e(H) + s(n - c)$. Since $H \subseteq \widehat{H}$ and $d_H(v) \leq k + 1$ for every $v \in V(C)$, $2e(H) \leq (s - 1)s + (c - s + 1)(k + 1)$. Therefore $e(G) \leq s(n - c) + \lfloor \frac{(s-1)s + (c-s+1)(k+1)}{2} \rfloor \leq F_k(n)$ by Lemma 3.9(i), contradicting Lemma 5.4(1). We may therefore assume that Lemma 4.4 (ii) holds. Then there is a set $R \subseteq V(C)$ such that $|R| = \lfloor \frac{c}{2} \rfloor - 1$ and $d_{\widehat{H}}(u) \leq \lfloor \frac{c}{2} \rfloor$ for every $u \in R$. Since Theorem 4.2 applies when $c \geq 10$, we first distinguish the cases $c \leq 9$ and $c \geq 10$. For convenience, set $q = \lfloor c/2 \rfloor$.

Case 1. $c \leq 9$.

Since $c \geq 2\lfloor (k + 1)/2 \rfloor + 2$ by Lemma 5.4(5), we have $2 \leq k \leq 6$. Since $q \leq k + 1$, we have $2e(H) \leq (q - 1)q + (c - q + 1)(k + 1)$. By Lemma 4.8(ii), $e(G - C) + e_G(G - C, C) \leq q(n - c)$. It follows that

$$e(G) \leq q(n - c) + \left\lfloor \frac{(q - 1)q + (c - q + 1)(k + 1)}{2} \right\rfloor.$$

If $(k, n, c) \neq (3, 8, 6)$, then Lemma 3.10 gives $e(G) \leq F_k(n)$, a contradiction. It remains to consider $(k, n, c) = (3, 8, 6)$. Then $e(G) = F_3(8) + 1 = 17$. Moreover, $|R| = 2$, $d_{\widehat{H}}(u) \leq 3$ for $u \in R$, and Lemma 5.3 gives $d_H(x) \leq 4$ for $x \in V(C)$. Hence $2e(H) \leq 2 \cdot 3 + 4 \cdot 4 = 22$ and $e(H) \leq 11$. By Lemma 4.8(ii), $e(G - C) + e_G(G - C, C) \leq 3(8 - 6) = 6$. Consequently,

$$e(G) = 17, \quad e(H) = 11, \quad e(G - C) + e_G(G - C, C) = 6. \quad (5.1)$$

By Lemma 5.4(4), $\delta(G) \geq 3$. Let $V(G) \setminus V(C) = \{u, v\}$. We claim that $uv \notin E(G)$. Otherwise, we have $|N_C(u)|, |N_C(v)| \geq 2$ and (5.1) gives $|N_C(u)| + |N_C(v)| = 5$. We may assume that $|N_C(u)| = 3$. For each $y \in N_C(v)$, there is an $x \in N_C(u) \setminus \{y\}$ at distance at most two from y on C . Since $xuvy$ is a path of length 3, replacing the shorter segment xCy by $xuvy$ gives a cycle of length at least 7, a contradiction. Thus, $uv \notin E(G)$ and $|N_C(u)| + |N_C(v)| = 6$. Since $\delta(G) \geq 3$, it follows that $|N_C(u)| = |N_C(v)| = 3$. Let $C = a_1b_1a_2b_2a_3b_3a_1$. Since C is a longest cycle of G , neither $N_C(u)$ nor $N_C(v)$ contains two consecutive vertices of C . Hence $N_C(u)$ and $N_C(v)$ are the two independent sets of order 3 in C . If $N_C(u) = \{a_1, a_2, a_3\}$ and $N_C(v) = \{b_1, b_2, b_3\}$, then $ua_1b_1a_2b_2vb_3a_3u$ is an 8-cycle, a contradiction. Thus $N_C(u) = N_C(v)$. Without loss of generality, assume that $N_C(u) = N_C(v) = \{a_1, a_2, a_3\}$. If, say, $b_1b_2 \in E(G)$, then $ua_1b_3a_3b_2b_1a_2u$ is a 7-cycle, a contradiction.

By (5.1), the cycle C has five chords. Since no chord joins two vertices of $\{b_1, b_2, b_3\}$, every chord has one of the following types:

- Type I: an edge inside $\{a_1, a_2, a_3\}$ (at most 3);
- Type II: a_1b_2 , a_2b_3 and a_3b_1 .

Let m be the number of type-I chords. Since there are at most three type-II chords, $m \geq 2$.

In H , every a_i has degree at most 4 and is already incident with two edges of the cycle C . Hence, each a_i is incident with at most two chords. Therefore, there are at most six chord incidences at a_1, a_2, a_3 . On the other hand, each type-I chord contributes two such incidences, while each type-II chord contributes one. Since there are m type-I chords and $5 - m$ type-II chords, the total number of chord incidences is $2m + (5 - m) = m + 5 \geq 7$, a contradiction.

Case 2. $c \geq 10$.

We distinguish according to whether $e(G) - e(H) \leq (q - 1)(n - c)$. In the second subcase, we use Theorem 4.2.

Subcase 1. $e(G) - e(H) \leq (q - 1)(n - c)$.

By Lemma 5.3, $d_H(v) \leq k + 1$ for every $v \in V(C)$. Hence

$$\begin{aligned} 2e(H) &= \sum_{v \in V(C)} d_H(v) = \sum_{u \in R} d_H(u) + \sum_{v \in V(C) \setminus R} d_H(v) \\ &\leq (q - 1) \min\{q, k + 1\} + (c - q + 1)(k + 1). \end{aligned}$$

By Lemma 3.9(ii), we have

$$e(G) \leq (q-1)(n-c) + \left\lfloor \frac{(q-1)\min\{q, k+1\} + (c-q+1)(k+1)}{2} \right\rfloor \leq F_k(n),$$

contradicting Lemma 5.4(1).

Subcase 2. $e(G) - e(H) > (q-1)(n-c)$.

We begin with analyzing the structure of G .

Claim 5.2. $G \subseteq W_{n,q,c}$.

Proof. Suppose not. By Lemma 5.4, G is 2-connected and $c < n$. Since $c \geq 10$, Theorem 4.2 applies. Then c is odd and $G \subseteq F$ for some $F \in \mathcal{X}_{n,c} \cup \mathcal{Y}_{n,c}$. If $k \leq 3$, then $n \leq 8$, contradicting $c \geq 10$. If $k = 4$, then $n \leq 11$ and $10 \leq c < n$. Then $c = 10$, contradicting the fact that c is odd. Thus $k \geq 5$. Then $\delta(G) \geq \lfloor (k+1)/2 \rfloor + 1 \geq 4$. By Lemma 4.3, either $F \subseteq W_{n,q,c}$ or F has a vertex of degree at most 3. If $F \subseteq W_{n,q,c}$, then $G \subseteq W_{n,q,c}$. If $\delta(F) \leq 3$, then $\delta(G) \leq 3$, a contradiction. Hence $G \subseteq W_{n,q,c}$. $\square$

Since $G \subseteq W_{n,q,c}$, $V(G)$ can be partitioned into three disjoint vertex sets S, T, U with $|S| = q$ and $|T| = \tau := c - 2q + 1 \in \{1, 2\}$.

Claim 5.3. $S \subseteq V(C)$ and $|T \setminus V(C)| \leq 1$.

Proof. Orient C and let u^+ be the successor of u on C . If $u \in U \cap V(C)$, then $u^+ \in S \cap V(C)$ since the vertices in U are adjacent only to the vertices in S . Since the successor map is injective, $|U \cap V(C)| \leq |S \cap V(C)|$. Hence, we have $c \leq 2|S \cap V(C)| + |T \cap V(C)| \leq 2|S \cap V(C)| + \tau$. If $S \not\subseteq V(C)$, then $|S \cap V(C)| \leq q-1$. Therefore $c \leq 2(q-1) + \tau = c-1$, a contradiction. Thus $S \subseteq V(C)$. Now $c \leq 2q + |T \cap V(C)|$, while $c = 2q + \tau - 1$. It follows that $|T \cap V(C)| \geq \tau - 1$. Hence $|T \setminus V(C)| \leq 1$. $\square$

By Claim 5.3, we have $S \subseteq V(C)$ and $|T \setminus V(C)| \leq 1$. Since U is independent, $E_G(U, T) = \emptyset$ and $|T \setminus V(C)| \leq 1$, the graph $G - C$ has no edges. Thus $e(G) - e(H) = e_G(G - C, C)$.

We first prove that $q \in I_k$. Since $e(G) - e(H) \geq (q-1)(n-c) + 1$, there exists a vertex $z \in V(G) \setminus V(C)$ with at least q neighbors on C . If $q \geq k+2$, then there exists a subpath P on C that contains at least $k+2$ neighbors of z . Thus, z and P form a $(k+2)$ -path-fan, a contradiction. Therefore $q \leq k+1$. On the other hand, Lemma 5.4 gives $c \geq 2\lfloor (k+1)/2 \rfloor + 2$. Then, $q \geq \lfloor (k+1)/2 \rfloor + 1$. Hence $q \in I_k$.

Now

$$e(G) - e(H) = \sum_{v \in U \setminus V(C)} |N_G(v) \cap V(C)| + \sum_{v \in T \setminus V(C)} |N_G(v) \cap V(C)|.$$

Set $\tau_C = |T \cap V(C)|$. Then $|T \setminus V(C)| = \tau - \tau_C$ and $\tau_C \in \{\tau-1, \tau\}$. If $v \in U \setminus V(C)$, then all its neighbors are in S . Thus, $|N_G(v) \cap V(C)| \leq q$. If $v \in T \setminus V(C)$, then its neighbors on C are in $S \cup (T \cap V(C))$. Hence $|N_G(v) \cap V(C)| \leq q + \tau_C$. Since $|U \setminus V(C)| = (n-c) - (\tau - \tau_C)$, we obtain

$$e(G) - e(H) \leq q(n-c - (\tau - \tau_C)) + (q + \tau_C)(\tau - \tau_C) = q(n-c) + \tau_C(\tau - \tau_C). \quad (5.2)$$

Let $X = V(C) \setminus S = (T \cup U) \cap V(C)$. Then $|X| = c - q$. Since U is independent and has no neighbors in T , we have

$$e(H) \leq e(G[S]) + e_G(S, X) + \binom{\tau_C}{2}. \quad (5.3)$$

Since $\tau \in \{1, 2\}$ and $\tau_C \in \{\tau - 1, \tau\}$, we have $\binom{\tau_C}{2} + \tau_C(\tau - \tau_C) = \tau - 1 \in \{0, 1\}$. By (5.2) and (5.3),

$$\begin{aligned} e(G) &\leq e_G(S, X) + e(G[S]) + \binom{\tau_C}{2} + q(n - c) + \tau_C(\tau - \tau_C) \\ &\leq e_G(S, X) + e(G[S]) + q(n - c) + \tau - 1. \end{aligned} \quad (5.4)$$

Applying Lemma 5.3 to the vertices of $S \subseteq V(C)$ gives

$$2e(G[S]) + e_G(S, X) = \sum_{s \in S} d_H(s) \leq q(k + 1). \quad (5.5)$$

It remains to prove that $e(G) \leq J_k(n)$. First, suppose that $|X| = c - q \leq k + 1$. Thus, by (5.5), $e(G[S]) \leq \lfloor \frac{q(k+1) - e_G(S, X)}{2} \rfloor$. Therefore,

$$e(G) \leq e_G(S, X) + \left\lfloor \frac{q(k+1) - e_G(S, X)}{2} \right\rfloor + q(n - c) + \tau - 1.$$

Since $e_G(S, X) \leq q(c - q)$,

$$\begin{aligned} e(G) &\leq q(c - q) + \left\lfloor \frac{q(k+1) - q(c - q)}{2} \right\rfloor + q(n - c) + \tau - 1 \\ &= q(n - q) + \left\lfloor \frac{q(k+1 - c + q)}{2} \right\rfloor + \tau - 1. \end{aligned}$$

Since $c = 2q + \tau - 1$, $k + 1 - c + q = k + 2 - q - \tau$. Hence

$$e(G) \leq q(n - q) + \left\lfloor \frac{q(k+2 - q - \tau)}{2} \right\rfloor + \tau - 1.$$

If $\tau = 1$, then $e(G) \leq q(n - q) + \left\lfloor \frac{q(k+1 - q)}{2} \right\rfloor \leq J_k(n)$. If $\tau = 2$, then $c = 2q + 1$. Since $c - q \leq k + 1$, $q \leq k$. Then,

$$\begin{aligned} e(G) &\leq q(n - q) + \left\lfloor \frac{q(k - q)}{2} \right\rfloor + 1 = q(n - q) + \left\lfloor \frac{q(k - q)}{2} + 1 \right\rfloor \\ &\leq q(n - q) + \left\lfloor \frac{q(k + 1 - q)}{2} \right\rfloor \leq J_k(n). \end{aligned}$$

Next, we assume that $c - q > k + 1$. Since $c = 2q + \tau - 1$ and $\tau \in \{1, 2\}$, if $\tau = 1$, then $c - q = q \leq k + 1$, a contradiction. Hence $\tau = 2$. Then, $c - q = q + 1$ and $q + 1 > k + 1$, that is, $q > k$. Since $q \leq k + 1$, $q = k + 1$. From (5.5), $2e(G[S]) + e_G(S, X) = \sum_{s \in S} d_H(s) \leq q(k + 1) = q^2$. Hence

$$e(G) \leq q^2 + q(n - c) + 1 = q^2 + q(n - 2q - 1) + 1 = q(n - q) - q + 1 \leq q(n - q).$$

Thus $e(G) \leq J_k(n)$.

Therefore, $e(G) \leq J_k(n) \leq F_k(n)$, contradicting $e(G) = F_k(n) + 1$. ■

The proof of Lemma 5.2 is very short.

Proof of Lemma 5.2. We proceed by induction on n . The base case $n = \lceil (5k + 1)/2 \rceil$ follows from Lemmas 5.1 and 3.7.

Let $n > \lceil (5k + 1)/2 \rceil$, and let G be an n -vertex graph containing no $(k + 2)$ -path-fan. Let $P = v_0 v_1 \dots v_t$ be a longest path. Then every neighbor of v_0 lies on P . If $d_G(v_0) \geq k + 2$, then v_0 and the path $v_1 v_2 \dots v_t$ form a $(k + 2)$ -path-fan, a contradiction. Hence $d_G(v_0) \leq k + 1$. By induction and Lemma 3.7, $e(G) = e(G - v_0) + d_G(v_0) \leq (k + 1)(n - k - 2) + (k + 1) = (k + 1)(n - k - 1)$. This completes the proof. ■

Now, we can finish the proof of Theorem 1.3.

Proof of Theorem 1.3. If $k = 1$, then $g_1(3) = 3$. Pósa's theorem and the construction $K_{2,n-2}$ give $g_1(n) = 2n - 4$ for $n \geq 4$. We may therefore assume that $k \geq 2$. If $k + 2 \leq n < \lceil (5k + 1)/2 \rceil$, then Lemma 5.1 gives $g_k(n) \leq F_k(n)$. The two constructions in Section 3 give $g_k(n) \geq R_k(n)$ and $g_k(n) \geq J_k(n)$. Thus, $g_k(n) = F_k(n)$.

Now let $n \geq \lceil (5k + 1)/2 \rceil$. Lemma 5.2 gives $g_k(n) \leq (k + 1)(n - k - 1) = J_k(n)$. Construction 3.3 with $a = k + 1$ gives a graph containing no $(k + 2)$ -path-fan and having exactly $(k + 1)(n - k - 1)$ edges. Thus, $g_k(n) = J_k(n) = (k + 1)(n - k - 1)$. By Lemma 3.7, $F_k(n) = J_k(n)$. This completes the proof. ■

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Declaration of AI usage

This work began while the authors were using GPT-5.5 Pro. Early on, the AI system could not provide useful mathematical assistance. The first author proved several small- k ($k \leq 7$), and the second author proved a stability version of Jiang's theorem, relying heavily on techniques from his earlier joint paper with Ma [18].

The second author then combined the two manuscripts and provided them to the AI, asking it to propose a general conjecture. Its first proposed conjecture was incorrect; after further exchanges, its second conjecture was given and became the main theorem of this paper. Thus, the AI contributed to the formulation of the general conjecture, based on mathematical results and proofs already obtained by the authors. The authors also used AI for some calculations in auxiliary lemmas and for language editing and proofreading.

The current version was completely written and checked by the authors, who take full responsibility for the mathematical content and the final text. All original mathematical results proved in this paper, including all lemmas and claims proved in Sections 3–5 as well as the main theorem, Theorem 1.3, have been formalized and machine-checked in Lean 4.

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